Detecting and Analyzing Biases in Age Estimation Models: A Comprehensive Evaluation of Datasets and their biases*

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***Abstract*—Biases within facial age estimation datasets significantly impact model accuracy and fairness, raising critical concerns for responsible AI deployment across diverse demographic groups. This research will uncover and address the critical issue of dataset biases in facial age estimation models, focusing on the impact these biases can have on model performance and fairness across diverse demographic groups. This study employs a quantitative approach to comprehensively identify and quantify biases in facial age estimation datasets and models. By analyzing multiple facial image datasets and implementing fairness evaluation techniques after model implementation, this research seeks to enhance understanding of biases in age estimation models. The findings will contribute to the development of more transparent AI systems and provide guidelines for improving dataset documentation and fairness evaluation in the field of computer vision**

***Index Terms*—Age Estimation, Dataset Bias, Fairness Metrics, Bias Detection, Computer Vision, AI transparency**

I. Introduction

The field of facial age estimation has gained significant interest across various domains, with particular relevance in biometric applications [7], [12], [13]. Developments in this field have allowed for the rise of a plethora of datasets employed for a wide range of tasks. However, The accuracy and fairness of age estimation models are often compromised by hidden biases in training datasets, resulting in significant variations in model performance across race and gender groups [5]. In addition to the issue of how bias affects model performance, the lack of proper documentation in facial image datasets contributes to hidden biases. And the emphasis on overall performance often obscures the nature of the biases produced by the actual fairness issues. Making many implementation ultimately biased. Moreover, the biases in facial age perception are often more pronounced in AI systems then in human judgments. [7].This research focuses on detecting and analyzing these dataset and model biases to improve understanding of fairness issues in age estimation models.

II. Research Methodology

This study employs a quantitative approach to comprehensively examine age, gender and racial biases with regards to facial age estimation.

A. Documenting Datasets and Classifying Biases

To ensure transparency and structure dataset documentation the "Datasheets for Datasets" framework is used [2]. This provides a general framework for improving dataset documentation and transparency. This approach has been analyzed and applied to discover it's specific usefulness in aiding the process of determining biases.

Additionally, for interpreting various types of biases, Mehrabi et al. [4] offers a comprehensive taxonomy of various types of bias in ML. Applying this taxonomy to the performed analysis helps with creating a nuanced understanding of the biases that are present in each system including their sources and manifestations.

B. Dataset and model Selection

To determine the biases prevalent in facial image datasets the demographic distributions of several datasets have been preliminarily analyzed on initial perceived diversity, initial bias/fairness remarks, realistic variability and the source which the data comes from (see table: I). The datasets that have been chose from this initial selection are the Fairface [15] and UTKface [14] datasets. FairFace [15] was explicitly created to provide balanced representation across race, gender, and age categories, making it valuable for bias measurement and mitigation. UTKFace [14], considered the industry standard in age estimation research, offers broad coverage and diversity. Together, these datasets enable comprehensive demographic analysis critical for evaluating model fairness.

The ResNet-50 [16] model was selected for cross-validation due to its proven ability in facial feature extraction and recognition tasks. ResNet-50's architecture enables effective generalization across different demographics, making it an appropriate choice to evaluate bias across age and race subgroups within these datasets.

C. Exploratory Data Analysis (EDA)

For each dataset, analysis have been conducted on the following fields:

- Age distribution,
- Gender distribution,
- Ethnic distribution,
- Completeness of annotations,
- Consistency of labeling across datasets(e.g how ethnicity's are grouped)

To evaluate age estimation performance while also analyzing race and gender biases, the data is partitioned and analyzed on different groups of interest for analyzing the biases (in particular gender and race). The images are separated based on the available image annotations, therefore it is found that the analysis performed is heavily reliant on the completeness of annotations within the dataset. Some features are considered complete quite easily (0 for men and 1 for females adequatly indicates gender), for other features this is a lot more difficult. The UTKFace dataset provides 5 race labels(White,Black,Asian,Indian and Others), while the FairFace dataset has 7 race labels (East Asian,Indian,Black,White,Middle Eastern,Latino Hispanic and Southeast Asian. There is no single definite classification of human races. This is further visualized in Fig. 1.

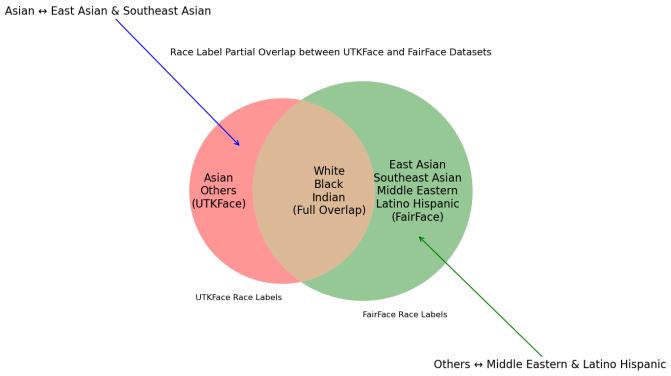


Fig. 1. Race label inter-dataset overlap

D. Fairness Metrics Implementation

To address these challenges, this research will use the Fairlearn library [1], an open-source tool designed to assess and improve the fairness of AI systems. By incorporating Fairlearn's capabilities, this study aims to quantify biases in age estimation models, ensuring construct validity through carefully selected metrics. This will be done using the following two complementary fairness metrics:

- Demographic Parity: This metric will help measure whether the age estimation model's age predictions are independent of ethnicity. This is done by comparing the likelihood that individuals from different ethnic groups receive the same prediction outcome, regardless of their actual age.
  Mathematically, demographic parity is expressed as:
  $$P(\hat{Y} = 1|A = a) = P(\hat{Y} = 1|A = b) \quad \forall a, b \in A \tag{1}$$

  Where:
  - $\hat{Y}$ is the predicted outcome (estimated age)
  - A is the sensitive attribute (e.g., race)
  - a, b are different values of the sensitive attribute (e.g. race labels)
- Equalized Odds: This metric evaluates whether the model's true positive and false positive rates are similar across different demographic groups. This means that individuals from different demographic groups should have similar chances of being correctly or incorrectly classified at the group level.
  Mathematically, equalized odds is expressed as:
  $$P(\hat{Y} = 1|A = a, Y = y) = P(\hat{Y} = 1|Y = y) \quad \forall a \in A, y \in \{0, 1\} \tag{2}$$

  Where:
  - $\hat{Y}$ is the predicted outcome
  - Y is the actual outcome
  - A is the sensitive attribute (e.g., race)
  - a represents different values of the sensitive attribute
  - $y \in \{0, 1\}$ represents binary outcomes

Each dataset was individually analyzed to identify inherent biases. Using a ResNet-50 model and applying the specified fairness metrics, this study systematically identifies and evaluates these biases. The analysis is structured as follows:

The model is kept as simple as possible to ensure clarity in approach. a pre-trained (ResNet-50) architecture was used to leverage existing knowledge while avoiding unnecessary complexity. The model was loaded with pre-trained weights, excluding the top layer to allow for customization tailored to my specific classification task of predicting age.

The model is then fitted using the following hyperparameters:

- Optimizer: Adam (learning rate = 0.001) is chosen for its adaptive step size, enabling efficient convergence across diverse age features.
- Loss Function: Categorical cross-entropy suits the task of age group classification by promoting accuracy in categorizing ages.
- Evaluation Metric: Accuracy provides a straightforward performance measure and, alongside fairness metrics, aids in assessing demographic balance.

E. Cross-Dataset Analysis

In order to gain more insight in the way that biases may transfer across datasets, the ResNet-50 model has been trained and evaluated on various individual datasets using the same methodology for analysis and framework for interpretation. Then the aforementioned fairness metrics have been used to see how these biases change when the same (pre-trained) models are applied to different datasets. By following this methodology, the aim is to provide a comprehensive analysis of biases in these facial age estimation datasets, contributing to improved transparency and fairness in age estimation models.

Validity and reliability in this study are addressed through careful use of fairness metrics, ensuring accurate assessment of bias. Validity is ensured by applying the model across different datasets. Consistent methodologies and clear documentation

will support the reliability of the results, allowing others to reproduce the research under similar conditions.

III. Results

A. Age Bias

Both datasets have an general over representation of certain age groups(20-29), making any model trained on the data likely to be biased toward these ages.

The UTKFace dataset shows a strong bias, with nearly 10% of images being labeled as age 26 and (almost 5%) as age 1. The latter may indicate annotation errors, suggesting unknown ages were defaulted to '1'. Further visual inspection of the data indeed revealed numerous mislabeled or unusable images. These findings highlight significant quality issues within the dataset, as visualized in Fig.3. While the FairFace dataset shows a bias towards age class 20-29 with 29,51% of the data being in this age group and more then half of the data being in the combined age group of 20-39. This is further visualized in Fig.2. To facilitate cross-dataset analysis the UTKFace age will be prepared for a classification problem using the same age groups as FairFace. When looking at Fig.4, it becomes apparent that UTKFace has no less bias toward the ages 20-29 then FairFace with 30.98% of the data being in this age group. These findings emphasize the importance of addressing dataset biases in age estimation models, as the uneven age distributions in both UTKFace and FairFace may introduce skewed predictions, impacting model reliability and fairness across demographic groups.

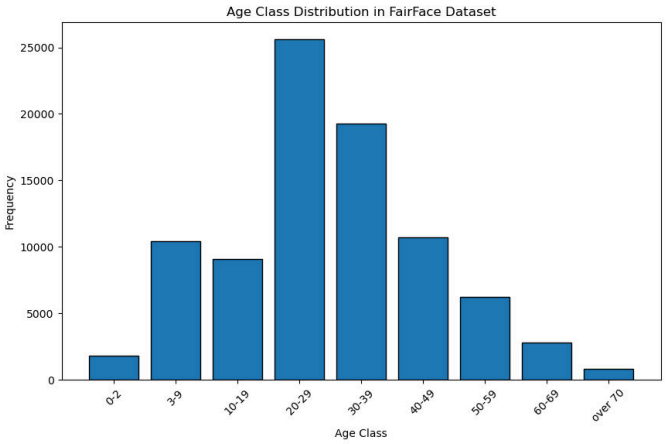


Fig. 2. Age Class Distribution in FairFace Dataset

B. Gender Bias

Both datasets have been equalized as far as gender goes, both have almost exactly the same gender distribution.(as seen in the appendix: (results analysis) therefore there is very little biases with regards to gender, the gender distribution across races however fluctuates highly in UTKFace, indicating for example a fair representation for women in general, but not necessarily the same representation for women of color(Fig.11).

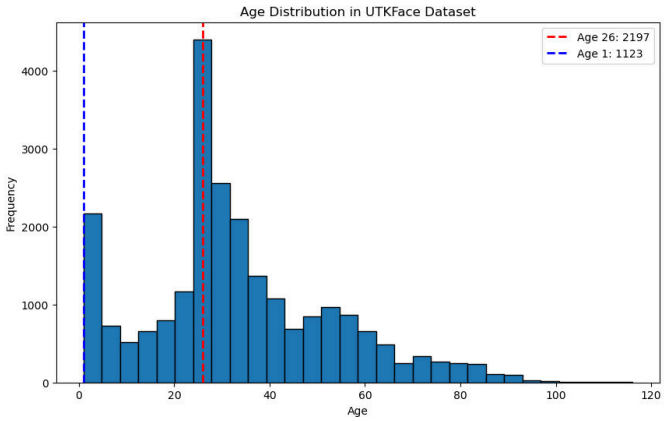


Fig. 3. Age Distribution in UTKFace Dataset

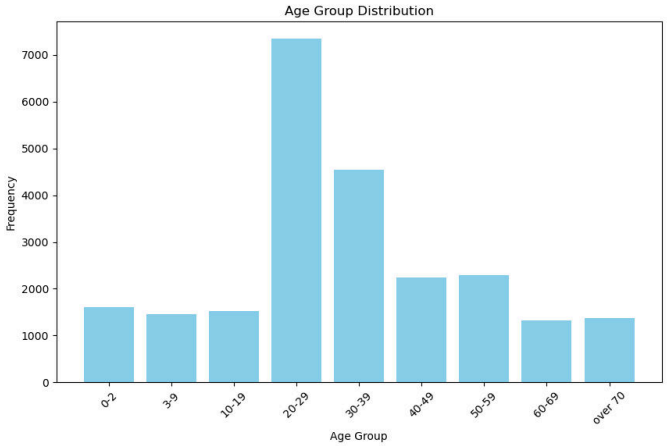


Fig. 4. UTK distribution with Fairface age groups

C. Racial Bias

1) Dataset Bias: When looking at Fig.5, it becomes apparent that the FairFace has a very racially equalized dataset, thus there is no considerable bias concerns on this regard, the biggest group is white, making up 19,1% of the dataset, which is only 4,8% more then the average ethnicity size in the dataset. this equality points to an effort of mitigating racial bias, promoting diversity. In contrast, UTKFace contains fewer race categories, with 'White' making up 42% of the dataset. (see: Fig.6). Additionally, UTKFace includes a vague 'Others' category, which is not only uninformative but also risks reinforcing biases by obscuring specific demographic information.

2) Model Bias: Figures: *8* and *7*, display the aforementioned fairness metrics as evaluated on model performance of the ResNet-50 model [16] on the two datasets. FairFace is shown to have higher fluctuations in Demographic Parity and Equalized Odds(see: Fig.8). The observed fluctuations in Demographic Parity and Equalized Odds suggest potential instability in model performance across racial groups.

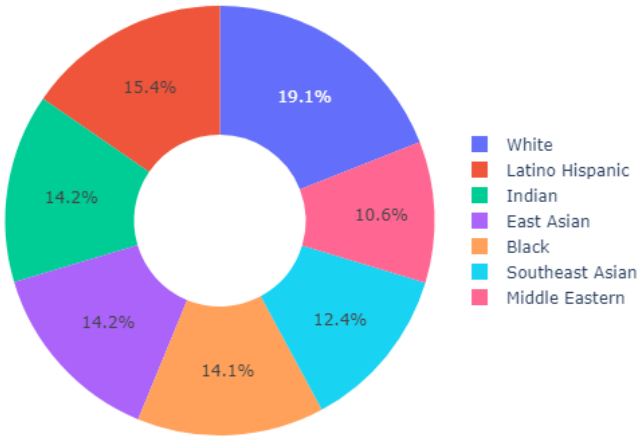


Fig. 5. Race Distribution of FairFace dataset

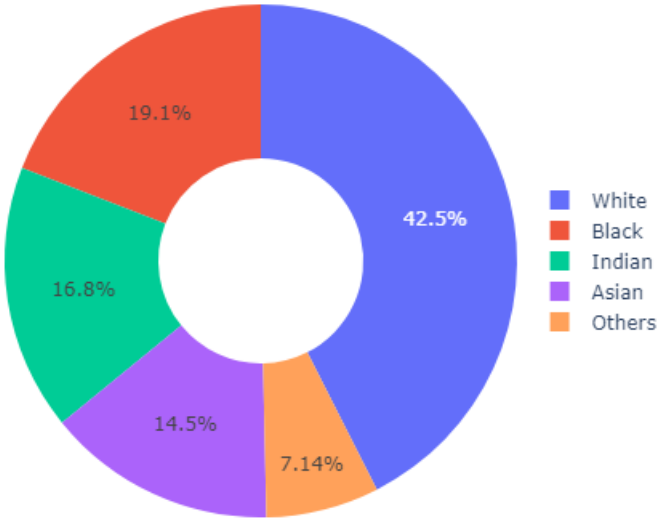


Fig. 6. Race Distribution of UTKFace Dataset

Conclusion

This study highlights the complexity of biases in facial age estimation models, revealing distinct performance characteristics when models are trained on the FairFace and UTKFace datasets. FairFace, designed for demographic balance, showed variability on the ResNet-50 model architecture in fairness metrics such as Demographic Parity and Equalized Odds, suggesting potential inconsistencies across racial groups. This instability could be linked to previous bias mitigation strategies that, while promoting representation of minorities, may have inadvertently introduced bias through annotation with a focus on (over)representing certain groups. In contrast, the simpler structure of UTKFace resulted in more stable model outcomes but constrained its ability to capture the demographic complexity necessary for a comprehensive evaluation of fairness in age estimation.

These findings underscore the importance of transparent

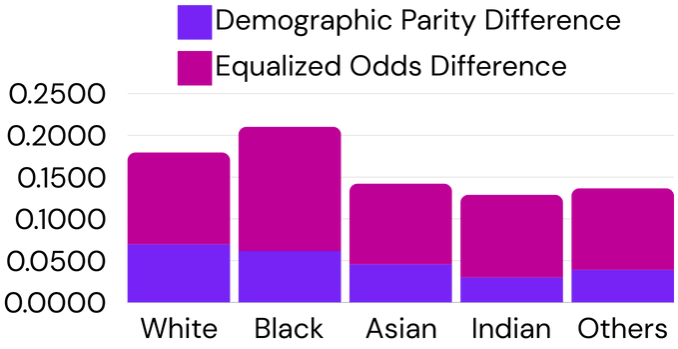


Fig. 7. Fairness metrics performance on UTKFace

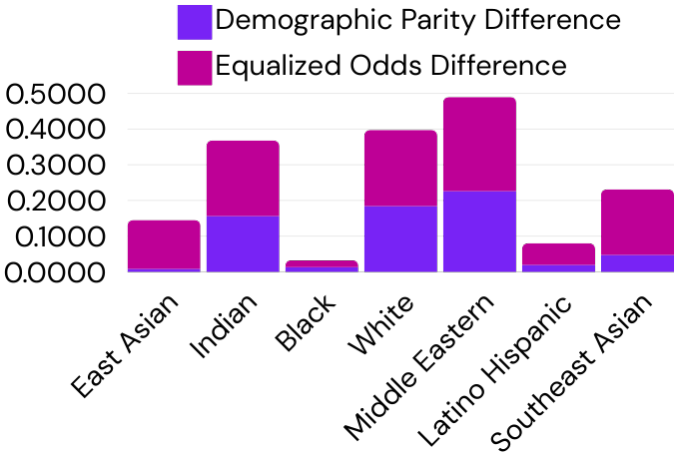


Fig. 8. Fairness metrics performance on FairFace

dataset documentation and the implementation of rigorous fairness evaluations to ensure that age estimation models deliver consistent and equitable results across diverse populations. Ensuring accountability in model fairness is essential as facial age estimation technology continues to advance.

Discussion

The model that is implemented to measure the biases in these dataset has not been optimized enough for real world implementation, this is because the methodology behind model architecture in this paper acts as a reference point as to not obscure the way that the predictions have been calculated. further hyper parameter tuning would be needed to optimize the performance at the cost of explainability and consistancy in cross-dataset Analysis.

In this research specific biases have been analyzed and documented, further research could look at the inner workings of bias sub-groups by additionally separating each race group by into multiple demographic sub-groups.

Future research could also further build on this study by incorporating multiple model architectures to enhance cross-validation efforts. By applying and comparing different models, a broader perspective on bias detection and mitigation across datasets could be achieved. This approach would also provide a more robust understanding of how biases persist

or vary across model architectures, ultimately contributing to more generalizable and fair age estimation systems.

APPENDIX

Appendix 1: Numbers from analysis

In this appendix I have included the results from the supporting analysis. For a full understanding of the analysis performed and how these results came about, please refer to the *full analysis on github*:

3) FairFace results:

a) Age class distributions:

- Percentage of records in age class 50-59: 7.18%
- Percentage of records in age class 30-39: 22.19%
- Percentage of records in age class 3-9: 12.00%
- Percentage of records in age class 20-29: 29.51%
- Percentage of records in age class 40-49: 12.39%
- Percentage of records in age class 10-19: 10.49%
- Percentage of records in age class 60-69: 3.20%
- Percentage of records in age class 0-2: 2.07%
- Percentage of records in age class over 70: 0.97%

b) Images per race:

- Number of East Asian: 12287
- Number of Indian: 12319
- Number of Black: 12233
- Number of White: 16527
- Number of Middle Eastern: 9216
- Number of Latino Hispanic: 13367
- Number of Southeast Asian: 10795

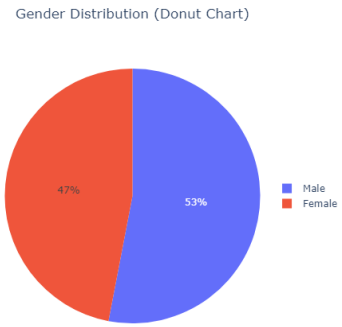


Fig. 9. Gender Distribution FairFace

4) UTKface results:

a) age class distributions:

- Percentage of records in age group 20-29: 30.98%
- Percentage of records in age group 30-39: 19.14%
- Percentage of records in age group 50-59: 9.70%
- Percentage of records in age group 40-49: 9.47%
- Percentage of records in age group 0-2: 6.77%
- Percentage of records in age group 10-19: 6.46%
- Percentage of records in age group 3-9: 6.15%
- Percentage of records in age group over 70: 5.79%
- Percentage of records in age group 60-69: 5.55%
- Percentage of records with age 1: 4.74%
- Percentage of records with age 26: 9.27%

b) Images per race:

- Number of White: 10078
- Number of Asian: 3434
- Number of Black: 4526
- Number of Indian: 3975
- Number of Others: 1692

c) gender amounts:

- Number of men: 12391
- Number of women: 11314

d) Fairness values:

- Average metrics for White: Average Demographic Parity Difference: 0.0695 Average Equalized Odds Difference: 0.1100
- Average metrics for Black: Average Demographic Parity Difference: 0.0615 Average Equalized Odds Difference: 0.1489
- Average metrics for Asian: Average Demographic Parity Difference: 0.0457 Average Equalized Odds Difference: 0.0965
- Average metrics for Indian: Average Demographic Parity Difference: 0.0300 Average Equalized Odds Difference: 0.0989
- Average metrics for Others: Average Demographic Parity Difference: 0.0389 Average Equalized Odds Difference: 0.0978

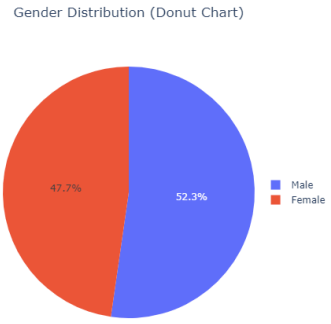


Fig. 10. Gender Distribution UTKFace

Appendix 2: Dataset selection

Dataset	Initial Perceived Diversity	Bias/Fairness	Realistic Variability	Comes from	Link
UTKFace	Moderate (Age, gender, ethnicity)	Very white and weird race labels	In the wild	Various (general web scrape)	link
FairFace	High (Race, gender)	Designed to be fair, balanced across demographics	In the wild	Yahoo YFCC100m dataset	link
Adience	Moderate (Limited to age and gender)	Limited age ranges (more young adults)	In the wild	Mainly flickr.com webscrape (automatic photo upload)	link
IMDB-WIKI	Low (Celebrity-focused)	Celebrity bias, less diversity (fewer elderly and maybe plastic surgery)	Controlled	IMDB and Wikipedia	link
CACD	Low (Celebrity-focused)	Celebrity bias, less diversity (fewer elderly and maybe plastic surgery)	Controlled	Celebs	link
MORPH	Low (probably a lot of young black men)	Probably biased toward African-American	Controlled	Mugshots	link
Lagenda	Moderate (Age, gender, no ethnicity label)	Seems very white	In the wild	Various (from openimages dataset)	link
FG-NET	Low (Seems mainly white)	Seems very white	In the wild	Family photos	link

TABLE I
COMPARISON OF DATASETS FOR AGE ESTIMATION AND BIAS ANALYSIS

supporting poster

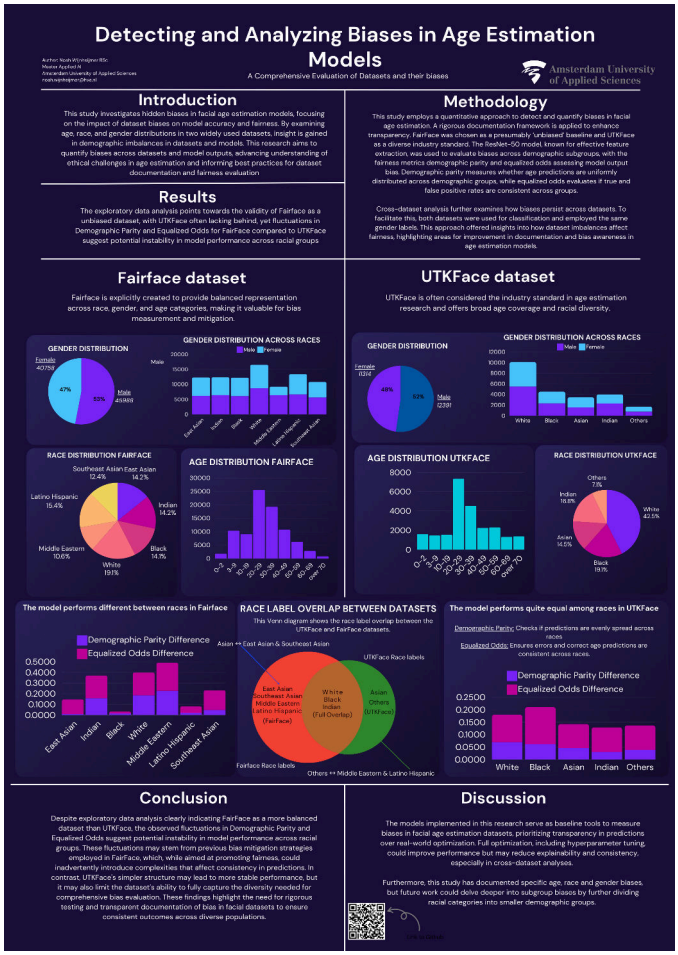


Fig. 11. Supporting poster for presentation purposes

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